



TEPCO Electricity Demand Forecasting

Deep Learning Implementation for Tokyo Power Grid Optimization.

View Portfolio



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Developed an end-to-end machine learning pipeline to predict hourly electricity demand in the Kanto region (Tokyo).

This project utilizes historical data (2020-2025) to facilitate better grid management and energy stability planning.

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Project Overview



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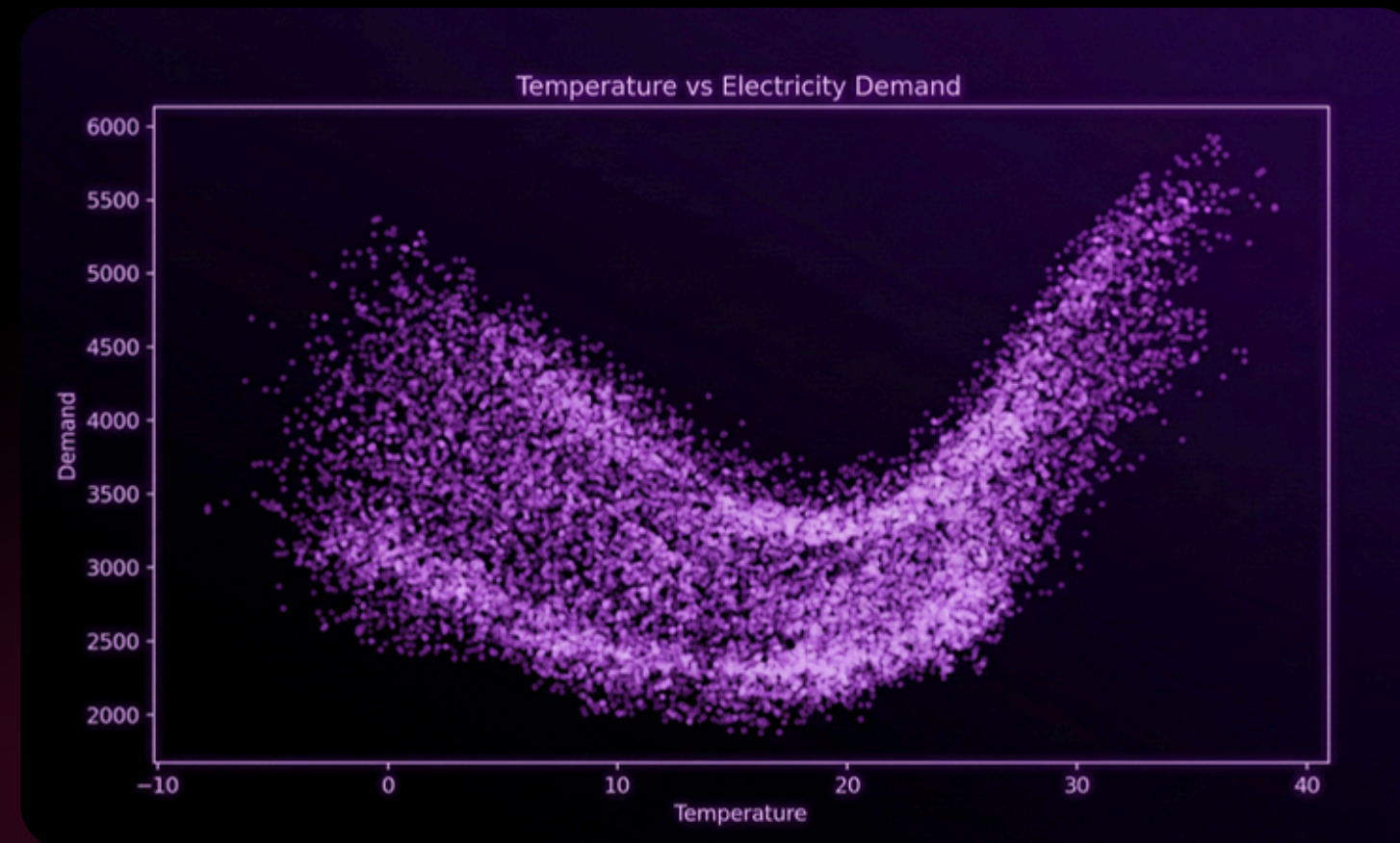
03



Key Data Insights

Exploratory Data Analysis (EDA) reveals a strong non-linear correlation between temperature and electricity consumption.

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Business Impact

Grid Stability: Accurate forecasting prevents blackouts by anticipating extreme load spikes in advance.

Cost Optimization: Enables utilities to optimize energy purchasing (reserve margin) at more efficient rates in the spot market.





End-to-End Pipeline

ETL Process

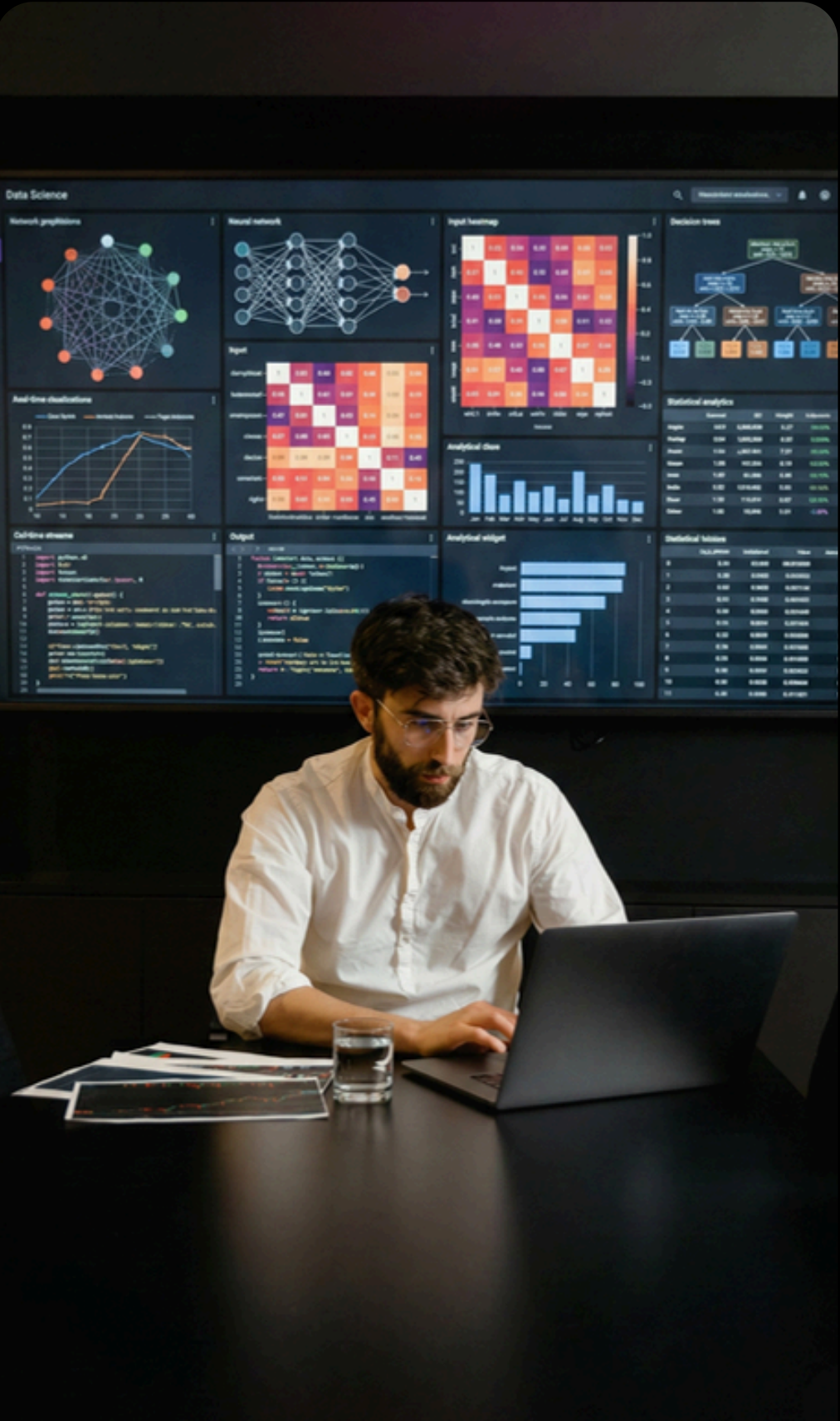
Parsed multi-year TEPCO raw data (CSV Shift-JIS) and integrated external weather data via Open-Meteo API.

Feature Engineering

Feature Engineering (Lag 1h/24h, Rolling Window) and Deep Learning Model Training (LSTM) using PyTorch.

Modeling

Trained Deep Learning models (LSTM) using PyTorch for high-precision time-series forecasting.



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Technology Stack

- Core: Python, Pandas, NumPy
- Modeling: PyTorch (LSTM), Facebook Prophet
- Viz & Dashboard: Streamlit, Plotly, Matplotlib
- Data: TEPCO Public Data, Open-Meteo API

Muhamad Juwandi

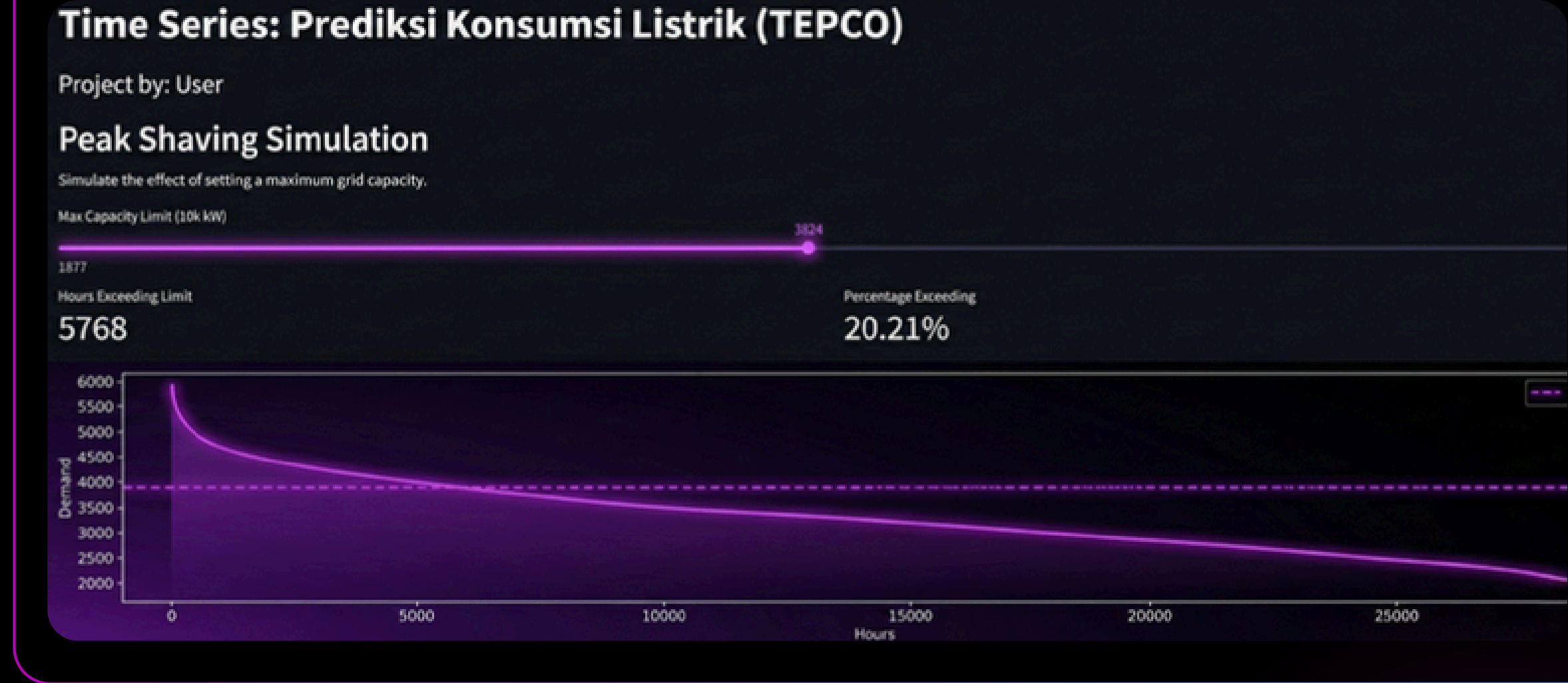
Home

Interactive Dashboard

A Streamlit-based dashboard enabling users to conduct load simulations and visualize model performance interactively.

2025 Data Science Project

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~150 MW

Rata-rata Error (MAE)

Model Performance



Baseline: Prophet



— Data Aktual — Prediksi Prophet

Effective in capturing annual seasonal trends, but lacks responsiveness to short-term fluctuations.

MAE ~250 MW

Deep Learning: LSTM



— Data Aktual — Prediksi LSTM

Superior performance in capturing complex non-linear patterns and short-term sequential dependencies.

MAE ~150 MW

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Future Improvements



Future roadmap includes implementing Temporal Fusion Transformer (TFT) for enhanced interpretability and deploying the model as a REST API using FastAPI.

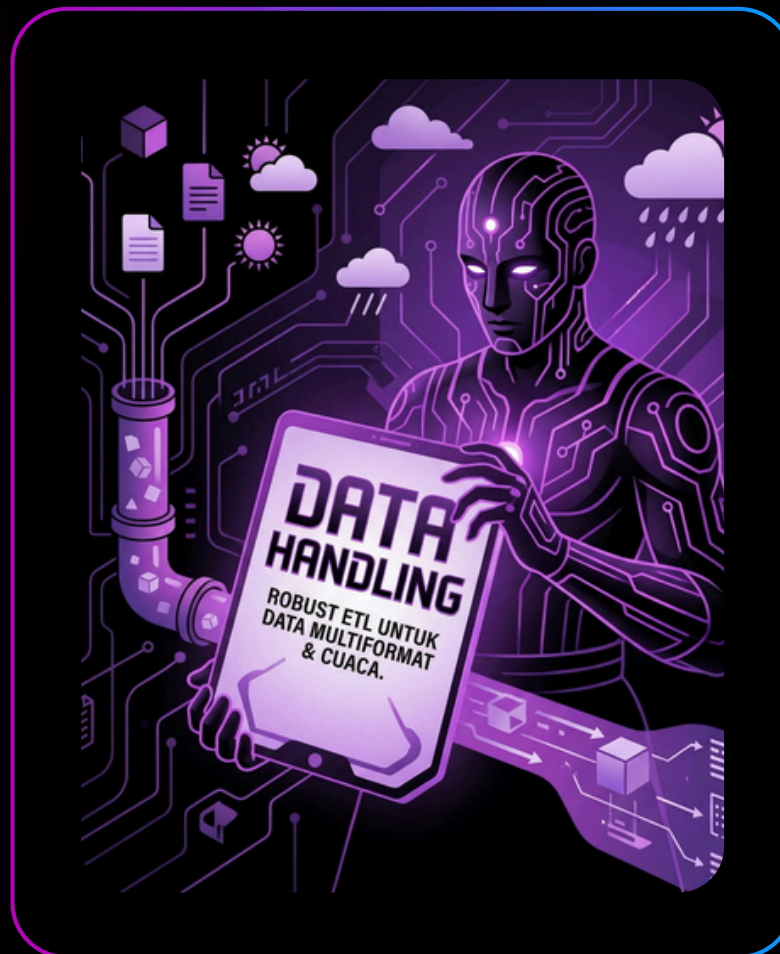
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Summary Strategy



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Let's Connect

Interested in discussing Time Series Forecasting or Machine Learning further? Reach out to me at:

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Thank You

FOR YOUR ATTENTION

Thank you for reviewing the TEPCO Demand Forecasting project portfolio. The complete source code is available in my GitHub repository.

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